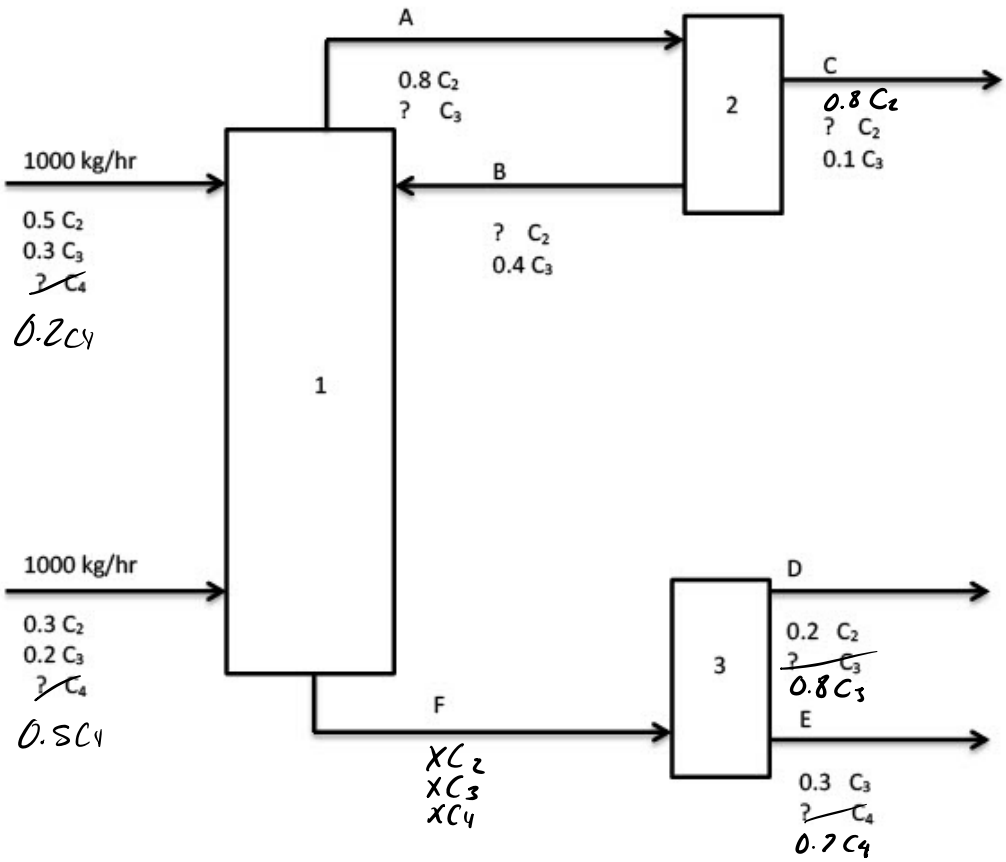
CHE 1411 L Group Project



Overall

in = out

Components:
C_2
C_3

$$\text{rref}\begin{bmatrix} 2000\,kg/h = C + D + E \\ 2000(0.3) = C(0.7) + D(0.2) + E(0) \\ 1000(0.3) + 1000(0.2) = C(0.1) + D(0.8) + E(0.3) \end{bmatrix}$$

$C = 820.5\ kg/h$
$D = 128.2\ kg/h$
$E = 1051.3\ kg/h$

Unit 2

Unit 2:
C_3:

$$\text{rref}\begin{bmatrix} C = A - B \\ 0.1C = 0.2A - 0.4B \end{bmatrix}$$

$A = 1231.5$

$B = 410.5$

Unit 3

$F = D + E + G$

$F = 128.2 + 1051.3 + 500$

$F = 1679.5$

		D	E	G
	in = out			
C_2:	$Fx_1 =$	$D(0.2)$ +	$E(0)$ +	$G(0)$
C_3:	$Fx_2 =$	$D(0.8)$ +	$E(0.3)$ +	$G(0.6)$
C_4:	$Fx_3 =$	$D(0)$ +	$E(0.7)$ +	$G(0.3)$

$x_1 = 0.02$
$x_2 = 0.43$
$x_3 = 0.53$

C_5: $x_4 = 1 - x_1 + x_2 + x_3$

$x_4 = 0.03$